

13장. 연습문제

In [1]:

```
#필요한 모듈 import
import warnings #출력창에서 경고문구를 무시

warnings.filterwarnings('ignore')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

np.random.seed(0)
```

In [2]:

```
# 데이터를 가져오기
df = pd.read_excel("사과-군집화.xlsx")
df.drop(columns=['kind', 'region'], inplace=True)
```

3번

In [3]:

```
import numpy as np
import pandas as pd

df = pd.read_excel("사과-군집화.xlsx")
df.drop(columns=['kind', 'region'], inplace=True)

from sklearn.cluster import KMeans

model1 = KMeans(n_clusters=5, random_state=0)
model1.fit(df)
```

Out[3]:

```
KMeans(n_clusters=5, random_state=0)
```

4번

In [4]:

```
import numpy as np
import pandas as pd
from sklearn.cluster import KMeans

df = pd.read_excel("사과-군집화.xlsx")
df.drop(columns=['kind', 'region'], inplace=True)
model1 = KMeans(n_clusters=5, random_state=0)
model1.fit(df)

df_c1 = df.copy()
df_c1['cluster'] = model1.labels_
df_c1
```

Out[4]:

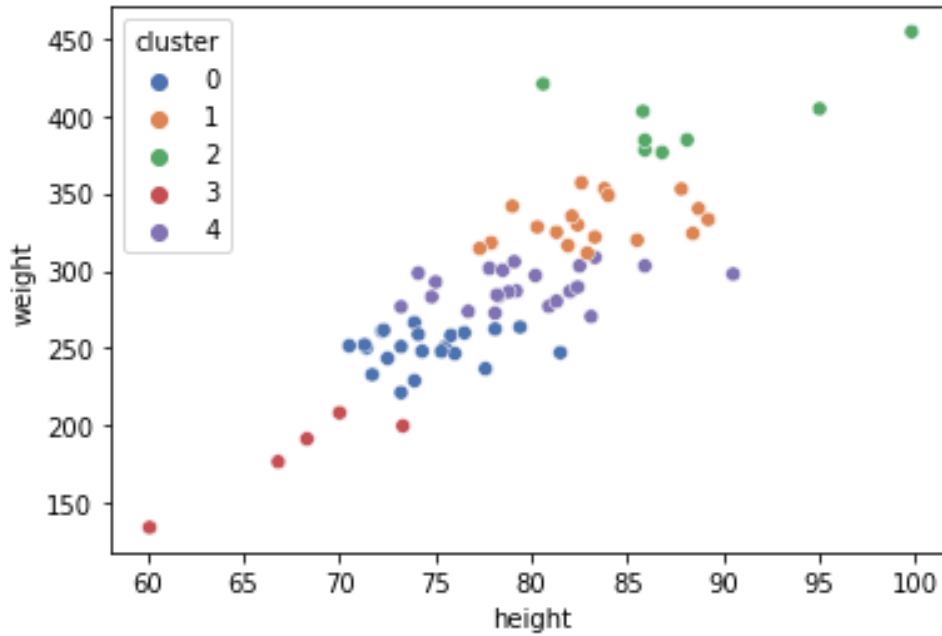
	weight	height	width	sugar	sour	cluster
0	283.6	78.3	87.0	13.0	0.39	4
1	229.0	73.9	81.6	11.4	0.29	0
2	266.5	73.9	86.1	12.3	0.37	0
3	176.7	66.8	73.7	11.0	0.31	3
4	308.6	83.3	90.7	15.2	0.19	4
...	...	...	...	...	...	...
71	384.2	85.9	96.4	14.1	0.51	2
72	311.3	82.9	90.5	12.7	0.25	1
73	284.3	78.2	88.0	13.6	0.36	4
74	324.0	88.4	92.1	12.6	0.22	1
75	273.7	76.7	85.6	12.6	0.29	4

76 rows × 6 columns

5번

In [5]:

```
sns.scatterplot(data = df_c1, x='height',y='weight',hue='cluster',palette='deep')
plt.show()
```



6번

In [6]:

```
from sklearn.metrics import silhouette_samples,silhouette_score

avg_score = silhouette_score(df, model1.labels_)
print('평균 실루엣 계수 =',avg_score)
```

평균 실루엣 계수 = 0.5292914473579787

In [7]:

```
print('평균 실루엣 계수 =', avg_score)
samples = silhouette_samples(df, model1.labels_)
score_df = pd.DataFrame({'cluster':model1.labels_, 'silhouette':samples})
score_df.groupby('cluster').describe()
```

평균 실루엣 계수 = 0.5292914473579787

Out[7]:

								silhouette
cluster	count	mean	std	min	25%	50%	75%	
0	22.0	0.601672	0.154277	0.235150	0.520720	0.647271	0.725827	0.74
1	18.0	0.508239	0.183265	0.020278	0.416558	0.557395	0.657845	0.68
2	8.0	0.536002	0.102015	0.372794	0.471253	0.552346	0.615244	0.65
3	5.0	0.464398	0.162378	0.198226	0.454564	0.481058	0.576531	0.61
4	23.0	0.488306	0.194997	0.074376	0.384353	0.547157	0.653698	0.69